A Comparative study of Black Box vs White Box models for Facial Recognition: Siamese Network and KNN

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1.Abstract

In the rapidly advancing field of digital security and identity verification, face recognition systems have become essential. This paper explores the comparative performance of blackbox and whitebox models for facial recognition using image similarity methods. We employ Siamese neural networks as blackbox models and K-Nearest Neighbors (KNN) as transparent glass box models, utilizing the Olivetti dataset for evaluation.

Siamese networks, with their ability to learn complex representations through a twin architecture, are effective in discerning subtle differences between faces by comparing high-dimensional embeddings. We initially implemented a custom Siamese network, which showed suboptimal performance. To enhance accuracy, we leveraged transfer learning with VGG19, significantly improving the model's efficacy. On the other hand, KNN, despite its simplicity and interpretability, faces challenges with high-dimensional facial data, impacting its ability to capture intricate patterns.

Our study highlights the robustness of Siamese networks, especially when combined with transfer learning, in handling limited data scenarios, contrasting with the straightforward yet less flexible nature of KNN. Grad-CAM was utilized for interpretability, providing insights into the decision-making process of the blackbox model, while saliency maps were used to enhance the transparency of the KNN model.

The findings underscore the trade-offs between model complexity, performance, and interpretability, emphasizing the need for tailored solutions in face verification systems. This research contributes to the ongoing discourse on the applicability of advanced neural architectures versus traditional machine learning methods in real-world security applications.

Keywords - Facial recognition , Image similarity check , Siamese Network , Few-shots learning , KNN

2.Introduction

In the evolving landscape of digital security and identity verification, the need for effective face verification and authentication systems has become paramount. This paper looks into the methodologies employed in face similarity checks, emphasizing the use of Siamese neural networks as black box models and contrasting them with transparent glass box models like K-Nearest Neighbors (KNN). Additionally, it highlights the rationale behind leveraging datasets such as the Olivetti dataset to address challenges in this domain.

Face verification and authentication systems are crucial in sectors like mobile device authentication, workplace attendance monitoring, and immigration checkpoints. With the advent of deep learning techniques, particularly neural networks, significant advancements have been made. Siamese neural networks, characterized by their twin architecture and ability to learn complex representations, have garnered attention for their efficacy in face similarity checks.

Siamese networks operate by comparing the similarity between two input images, making them ideal for tasks like face verification. By encoding facial features into high-dimensional representations, these networks can discern subtle differences between faces and accurately determine their similarity. This approach is particularly advantageous in scenarios with limited data availability, where traditional machine learning models may struggle to generalize effectively.

The choice of dataset is pivotal in developing and evaluating face verification systems.

The Olivetti dataset, comprising 400 face images of 40 individuals with 10 images per person, offers a compelling resource. Its controlled setting ensures consistency in image quality and facial expressions, facilitating robust model training and evaluation. Moreover, its modest size makes it conducive to exploring methodologies suited for few-shot learning, crucial in face verification systems. By leveraging this dataset, researchers can develop and test algorithms under realistic conditions, leading to more reliable and generalizable solutions.
In contrast to black box models like Siamese networks, glass box models like KNN offer transparency and interpretability. KNN operates by identifying the nearest neighbors based on a predefined distance metric, making it straightforward to understand and implement. However, its performance may be limited in high-dimensional data spaces, such as facial images, and it may struggle to capture complex patterns and variations.
Siamese networks are well-suited for face recognition tasks, designed to learn similarities between pairs of data points. This neural architecture, featuring twin subnetworks sharing weights, generates embeddings for input into a specialized loss function. This function minimizes the distance between similar inputs and maximizes the distance between dissimilar ones. Siamese networks offer significant advantages, such as effectively using small datasets by generating numerous image pairs, enhancing data efficiency, and allowing the addition of new classes without retraining. They reduce overfitting by focusing on relative differences, leading to better generalization. The contrastive loss function ensures similar images are embedded close together in the feature space, while dissimilar ones are far apart, enhancing accuracy in applications like security and authentication. Additionally, comparing embeddings is computationally efficient, making Siamese networks suitable for devices with limited resources. For this proposal, any face dataset with a small number of images can be used. The chosen dataset is the Olivetti dataset, which has a total of 400 face images for 40 people with 10 images per person. The face images were taken between April 1992 and April 1994, under varying lighting conditions, facial expressions, and facial details, with a black background. The images are gray level, sized at 64x64 pixels, and pixel values scaled to the [0, 1] interval. The names of the 40 people were encoded to integers from 0 to 39.

3.Literature Review

Face verification and authentication systems have become integral components of digital security and identity verification, attracting significant attention in both academic research and practical applications. This literature review offers a comprehensive examination of the methodologies, advancements, challenges, and future directions in this dynamic field. The literature on face verification spans a spectrum of methodologies, evolving from traditional machine learning approaches to cutting-edge deep learning techniques. Early methods relied on handcrafted features like Eigenfaces, Local Binary Patterns (LBP), and Histogram of Oriented Gradients (HOG) to represent and assess facial similarity. While effective to a degree, these methods struggled to capture the complexity of facial variations and adapt to diverse datasets.
The advancement of deep learning, particularly convolutional neural networks (CNNs), marked a significant breakthrough in face verification. CNN-based architectures such as Siamese networks, Triplet networks, and FaceNet revolutionized the field by directly learning hierarchical representations from raw pixel data. These models excel in discerning subtle facial features and generalize well to unseen faces, making them highly suitable for real-world applications.

3.1 Emerging tool in CNN/DeepLearning - Few-shot learning and Siamese Network

Traditional neural networks face limitations in tasks like facial recognition that require image similarity checks, primarily due to their dependency on large datasets for effective training and their tendency to struggle with generalization from limited data. This poses a challenge in real-world applications where obtaining extensive labeled datasets is impractical. Few-shot learning addresses this issue by enabling models to generalize well from a minimal number of examples. Siamese neural networks (SNNs) have emerged as a promising tool in this area due to their unique architecture, which learns to compare pairs of inputs rather than classifying single inputs. By using twin networks with shared weights, SNNs learn a similarity metric that can distinguish between similar and dissimilar image pairs, making them well-suited for few-shot learning scenarios. This capability is particularly beneficial for facial recognition, where SNNs can generate robust embeddings that accurately capture the features of different faces, even with limited training data. Key advancements like FaceNet and VGGFace have demonstrated the effectiveness of SNNs in achieving state-of-the-art performance in facial recognition by using techniques such as triplet loss and leveraging pre-trained models through transfer learning. Moreover, SNNs' adaptability to few-shot learning paradigms, such as prototypical networks and matching networks, further enhances their utility by improving their ability to recognize new classes with minimal examples. Despite their advantages, challenges remain, including handling variations in pose, lighting, and occlusions, as well as reducing computational costs. Ongoing research aims to address these issues and expand the applicability of SNNs, making them an increasingly vital tool for image similarity checks and facial recognition in artificial intelligence..

3.2 Emerging tool - Transfer Learning in Siamese Network

Another emerging tool in the field of artificial intelligence is transfer learning, which can be effectively used in conjunction with Siamese networks to overcome the limitations associated with using custom-developed base networks. Transfer learning involves leveraging a pre-trained model on a large dataset, such as ImageNet, to extract features that can be adapted to new tasks. This technique significantly reduces the amount of data and computational resources required to train deep learning models, while also improving their performance and generalization capabilities.

In the context of facial recognition, transfer learning can be particularly beneficial when integrated with Siamese networks. By utilizing a pre-trained model like VGG19, which has already learned rich feature representations from a vast dataset, the Siamese network can focus on learning the similarity metric between image pairs more effectively. This approach mitigates the challenge of training a custom base network from scratch, which often requires extensive computational power and large amounts of labeled data.

To illustrate, consider using VGG19 for transfer learning in a Siamese network designed for facial recognition. First, the VGG19 model is loaded with pre-trained weights, excluding its fully connected layers. The output of the last convolutional layer serves as a feature extractor, capturing essential facial features. These features are then fed into the Siamese network's twin architecture, which learns to compare the embeddings of two input images. By freezing the initial layers of VGG19 and fine-tuning only the top layers, the model can adapt the pre-trained features to the specific task of facial recognition, while retaining the robustness of VGG19's learned representations.

For example, in a practical application, the VGG19 model can be imported and modified to output 128-dimensional embeddings. These embeddings are used by the Siamese

network to calculate the similarity between facial images. This method not only enhances the efficiency and accuracy of the facial recognition system but also ensures that the model is robust and capable of handling various facial variations. By combining transfer learning with Siamese networks, researchers and practitioners can achieve high performance in facial recognition tasks with limited data and reduced training time.
In our task, using transfer learning with VGG19 improved the accuracy of the Siamese network when the custom model did not perform adequately. This approach allows for the utilization of robust, pre-trained feature extractors, thereby improving generalization and efficiency.

The literature suggests several avenues for future research and development in face verification. Continued advancements in deep learning techniques, including self-supervised learning and adversarial robustness, hold promise for improving system accuracy and robustness. Moreover, interdisciplinary collaborations integrating insights from computer vision, psychology, and sociology can enhance our understanding of human facial perception and inform the design of more effective verification algorithms.

3.3 Glassbox Models

Glass box models like K-Nearest Neighbors (KNN) offer transparent and interpretable approaches to face similarity assessment. Despite their simplicity and interpretability, KNN models may face challenges in high-dimensional data spaces like facial images due to the curse of dimensionality. Careful consideration of distance metrics and the number of neighbors selected is crucial to their performance.

Despite significant advancements, challenges persist in face verification, particularly concerning robustness and reliability in real-world scenarios. Factors such as varying lighting conditions, facial expressions, and occlusions can impact performance. Additionally, ethical considerations surrounding privacy, bias, and fairness must be carefully addressed to ensure responsible deployment of face verification systems.
Human-centered considerations are paramount in the development and deployment of face verification systems. Ethical considerations, including privacy, bias, and fairness, must be carefully addressed to ensure user trust and acceptance. Transparency and accountability are essential for fostering responsible deployment and adoption of face verification technologies..

4. Dataset and Data Exploration

As mentioned we are using the Olivetti dataset for this project which, consists of 400 images each of size 64x64, represents facial images of 40 individuals with 10 images per person (Figure 1). The dataset is readily available in sklearn datasets API. This dataset, along with its 400 corresponding labels, presents several characteristics and limitations that could introduce bias into facial recognition systems. A notable gender imbalance exists, with a majority of subjects being male, potentially leading to biased model performance skewed towards males. The age distribution is narrow, predominantly featuring young adults, which may hinder the model's ability to generalize to older adults or children. Additionally, the dataset lacks ethnic and racial diversity, with most subjects appearing to be of European descent, raising the risk of lower accuracy for individuals from different racial and ethnic backgrounds. The limited variations in

facial expressions, lighting, and poses further constrain the dataset's robustness.

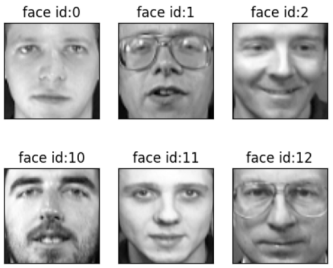


Figure 1: Olivetti Dataset Samples

These biases inherent in the Olivetti dataset—related to gender, age, and ethnicity—could lead to uneven model performance across different demographic groups. Deploying such a biased model could result in unfair treatment, misidentification, and discriminatory outcomes, perpetuating existing societal biases. While these issues are critical, their impact on our specific use case of image similarity check using Siamese Neural Networks or K-Nearest Neighbors (KNN) may differ, as these methods focus on comparing image pairs. However, the effects of dataset bias on image similarity checks are beyond the scope of this project and warrant future investigation. For preprocessing, the Siamese network requires pairs of images to learn similarities and dissimilarities. A function is created to generate positive and negative pairs: each image is paired with another image of the same label to form a positive pair, and with an image of a different label to form a negative pair. These paired images and their corresponding labels are then used as inputs for training the Siamese network.

5 Model Development:

5.1 Development of Blackbox Model

For Blackbox model we use Siamese Network as the model. Siamese Networks are well-suited for this task, as they are designed to learn similarities between pairs of data points, making them ideal for recognizing faces in different scenarios.This neural architecture is used for comparing the likeness or disparity between two input samples.It features twin subnetworks sharing weights (Figure 2), generating embeddings for input into a specialized loss function. This loss function minimizes the distance between similar inputs and maximizes the distance between dissimilar ones. So we will be using Siamese Network as our Blackbox model.

For the development of our black box model, we initially implement a Siamese network using a custom model. This means the twin subnet of the Siamese network will have a custome network structure. The definition of the custom defined base or backend model for our Siamese network is outlined below (Figure 3).

The model begins with an input layer that matches the shape of the input images. It then comprises three convolutional layers with increasing filter sizes, designed to capture spatial hierarchies and features at various levels of abstraction. The output from the convolutional layers is flattened to create a single long feature vector, which is subsequently fed into fully connected layers. A dense layer with 256 units and ReLU activation processes these flattened features, providing the final feature representation

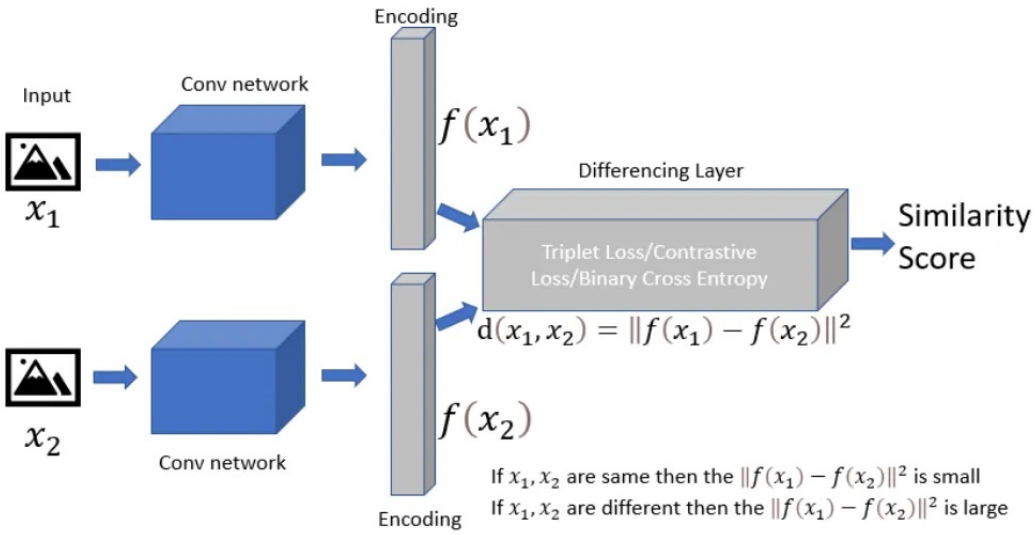


Figure 2: Architecture of Siamese Network

```
[ ]  # Define the base network (embedding model)
     def create_base_network(input_shape):
         input = Input(shape=input_shape)
         x = Conv2D(64, (7, 7), activation='relu')(input)
         x = MaxPooling2D()(x)
         x = Conv2D(128, (5, 5), activation='relu')(x)
         x = MaxPooling2D()(x)
         x = Conv2D(128, (3, 3), activation='relu')(x)
         x = Flatten()(x)
         x = Dense(256, activation='relu')(x)
         return Model(input, x, name='Embedding')
```

Figure 3: Base network of custom developed model for the Siamese network custom Model.

(embedding) of the input image. This function returns a Keras model named "Embedding" that encapsulates the defined layers and their connections.

In the Siamese network, two input layers represent the anchor and comparison images. Both input images are passed through the base network to obtain their embeddings, which are high-level feature representations of the images. The 'euclidean distance' function calculates the Euclidean distance between the embeddings of the two images, which is used to determine their similarity. The Distance layer computes this distance, and the Siamese network model is created by specifying its inputs and its output .

The model is then compiled using binary cross-entropy loss and the Adam optimizer. The binary cross-entropy loss is appropriate because the task is a binary classification (similar or dissimilar pairs). This setup ensures that the model is optimized to accurately distinguish between similar and dissimilar image pairs during training.

During training, the model is trained on the training data using pairs of images and their corresponding labels as inputs. The model is trained for 500 epochs with a batch size of 10, and 20 percentage of the data is used for validation.

5.2 Siamese Network with Transfer Learning

We go further in our development to make a Siamese Network with VGG19 as the Backbone or the base model of our network. We did this as we found the performance of the custom developed model (section 5.1) was not that great on Olivetti dataset, and we thought to develop another Siamese network with the hope that , a pretrained network with transfer learning could improve the accuracy (Figure 4).
VGG19 is a well-known deep learning model that has been pre-trained on a large dataset like ImageNet. This pre-training provides a strong feature extraction capability, which can be particularly useful for tasks like facial recognition. VGG19 has multiple convolutional layers that capture hierarchical features of images. These layers can extract intricate facial features, which are crucial for distinguishing between different individuals in the Olivetti dataset.
It's important to note that the Olivetti dataset consists of grayscale images, while VGG19 requires color images. Therefore, we need to convert the grayscale images to color explicitly.

```
# Load VGG19 with pre-trained ImageNet weights, excluding the top layer
base_network = VGG19(weights='imagenet', include_top=False, input_shape=(64, 64, 3))

# Freeze the layers in the base network
for layer in base_network.layers:
    layer.trainable = False

# Add a custom top layer
x = base_network.output
x = Flatten()(x)
x = Dense(256, activation='relu')(x)
```

```
Downloading data from https://storage.googleapis.com/tensorflow/keras-applications/vgg19/vgg19_weights_tf_dim_ordering_tf_ker
80134624/80134624 ━━━━━━━━━━━━━━━━━━━━ 0s 0us/step
```

```
from tensorflow.keras.layers import GlobalAveragePooling2D

# Create the Siamese network
input_a = Input(shape=(64, 64, 3), name='anchor')
input_b = Input(shape=(64, 64, 3), name='compare')

processed_a = base_network(input_a)
processed_b = base_network(input_b)

distance = Lambda(lambda embeddings: K.abs(embeddings[0] - embeddings[1]), name='distance_layer')([processed_a, processed_b])

# Add GlobalAveragePooling2D layer
x = GlobalAveragePooling2D()(distance)
# Add Dense layer for similarity prediction
x = Dense(1, activation='sigmoid')(x)
```

Figure 4: Siamese network with VGG19 as the base model.

5.3 Development of Glassbox Model

The development of the K-Nearest Neighbors (KNN) model for the face verification task, as part of the Siamese network approach, begins with preparing the dataset. The facial images are stored in 'pics' and corresponding labels in 'labels', which are then split into training and testing sets using 'traintestsplit', allocating 80 percentage for training and 20 percentage for testing. The KNN model is trained using the training set, with 'neighbors=1', meaning it considers only the closest neighbor for prediction. This simple and often effective approach works well for facial recognition tasks by utilizing a distance metric like Euclidean distance to classify test images based on the closest training samples. The model predicts labels for the test images, and its performance is evaluated by selecting random samples from the test set and comparing the true labels

with predicted labels. The images are plotted along with their true and predicted labels for visual inspection, providing insights into the model's accuracy. KNN offers simplicity and interpretability, but it requires careful selection of distance metrics and the number of neighbors to handle high-dimensional data effectively. Additionally, biases in the Olivetti dataset, such as gender imbalance and lack of diversity, can impact the model's reliability across different demographic groups. For more complex datasets, advanced techniques like Principal Component Analysis (PCA) or deep learning methods would be necessary to effectively reduce dimensionality and capture intricate patterns in the data.

6.Model Performance Evaluation

6.1 Blackbox Model

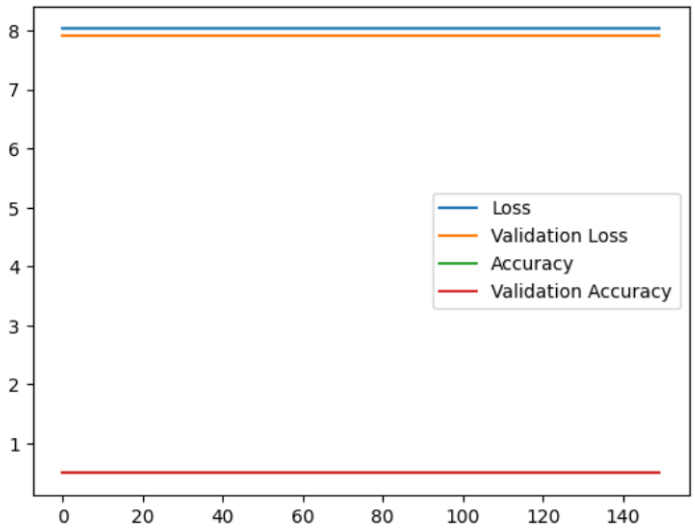


Figure 5: Model Performance of siamese network using Custom Model

The Siamese network was first developed with the custom base model (Section 5.1), and the model performance is not that great in that case , with accuracy of 49.61Due to this poor performance, we switched to VGG19 with Transfer Learning (Section 5.2).

Using VGG19 significantly improved the accuracy, boosting it to 95 percentage.Comparing the performance of the custom backbone model and the VGG19-based Siamese network highlights the significant advantages of using a pre-trained, sophisticated network architecture like VGG19 and transfer learning, over a simple custom model for facial recognition tasks.

So among the Blackbox models we got maximum better result with VGG19 trained Siamese network, and we will be considering that for our further anlaysis and comparison.

6.2 Model Performance of Glassbox Model

The glassbox model using KNN achieved impressive results, with an accuracy of 91.25 percentage. The precision of the model was 0.93, indicating a high proportion of true positive predictions among all positive predictions. The recall was 0.91, demonstrating the model's effectiveness in identifying true positives among all actual positives.

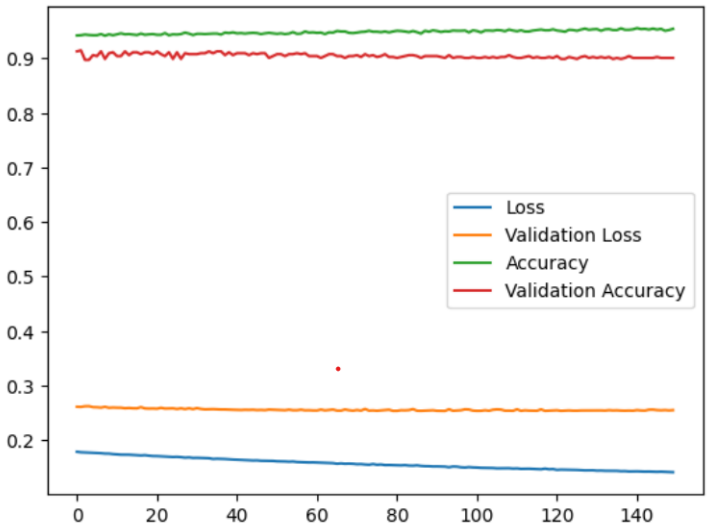


Figure 6: Model Performance of Siamese Model with VGG19

Additionally, the F1 score was 0.90, reflecting a good balance between precision and recall.

7.Explainability for the model

7.1 XAI for the Blackbox Model

To enhance the interpretability of our black-box model, we employed the Grad-CAM (Gradient-weighted Class Activation Mapping) technique, focusing on a Siamese network with VGG19 as the embedding model. One prepossessing step involved is modifying the grayscale Olivetti dataset images to color, as VGG19 requires color images. We then designed a function to generate Grad-CAM heatmaps for a specified convolutional layer. For our tests, we used the 'block5conv4' layer of the VGG19 model.

We applied Grad-CAM to pairs of similar and dissimilar images to visualize the areas of focus that contributed to the similarity scores. For the pair of similar images (Figure 7), we achieved a similarity score of 0.5. If we consider a threshold of threshold of 0.5, that the images were predicted as similar. The generated heatmaps highlighted the regions in both images that significantly influenced the model's decision. Overlaying these heatmaps on the original images provided a clear visual representation of the key features considered by the model.
In the case of dissimilar images, the similarity score was 0.62, suggesting that the images were predicted as not similar according to the same threshold. Again, the heatmaps and their overlays illustrated the critical regions that the model used to differentiate the images.

For a negative image pair shown in Figure 8, the similarity distance calculation gives score 0.62 , and if we keep a threshold of 0.5 then we can consider the images predicted as not similar. The visualization using Grad-CAM provided valuable insights into the decision-making process of our black-box model, making it more interpretable and understandable. The distinct areas highlighted in the heatmaps helped validate the model's performance and its focus on relevant features within the images.

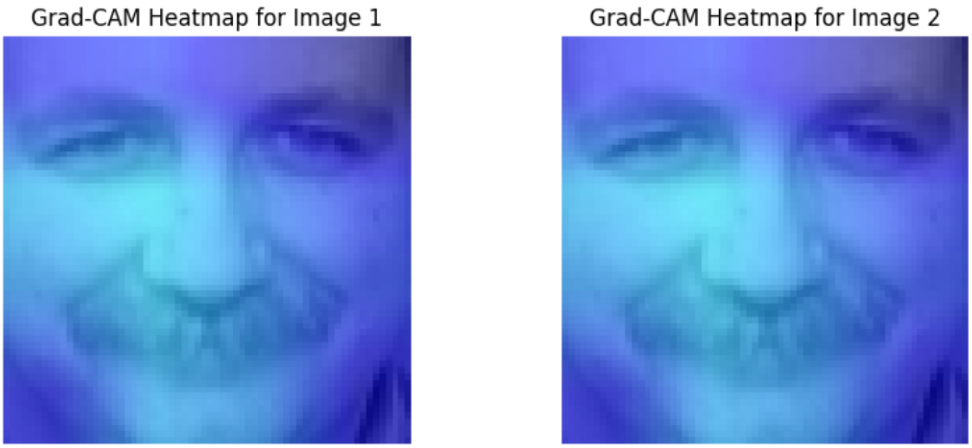


Figure 7: The Grad CAM visualization for positive pair Images

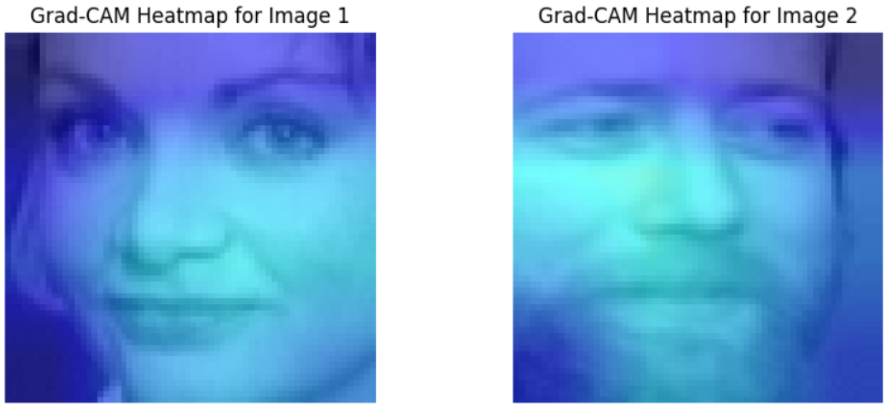


Figure 8: The Grad CAM visualization for negative pair Images

7.2 XAI for the Glassbox Model

To enhance the interpretability of our KNN-based glassbox model, we implemented several explainability techniques. We first generated saliency maps to visually highlight the impact of each pixel on the model's prediction through a perturbation process. By perturbing individual pixel intensities and observing changes in the model's output, we calculated pixel-wise perturbation scores. These scores were then reshaped into saliency maps, with regions of lower perturbation scores indicating pixels whose intensity changes had minimal effect on the prediction. We plotted the original test images alongside their saliency maps and overlayed heatmaps to provide a clear visual representation of influential pixels.
By computing the perturbation scores, we created saliency maps, which visually highlight the impact of each pixel on the model's prediction. In the saliency map, regions that appear black indicate pixels with low perturbation scores. This means that changes in the intensity of these pixels have little to no effect on the model's prediction.

Additionally, we attempted to use LIME (Local Interpretable Model-agnostic Explanations) to further explain the KNN model's predictions. By converting the grayscale test images to RGB format, we utilized the LimeImageExplainer to generate explanations for selected test images. However, this part of the code encountered a value error that prevented it from running successfully. The intention was to visualize the regions

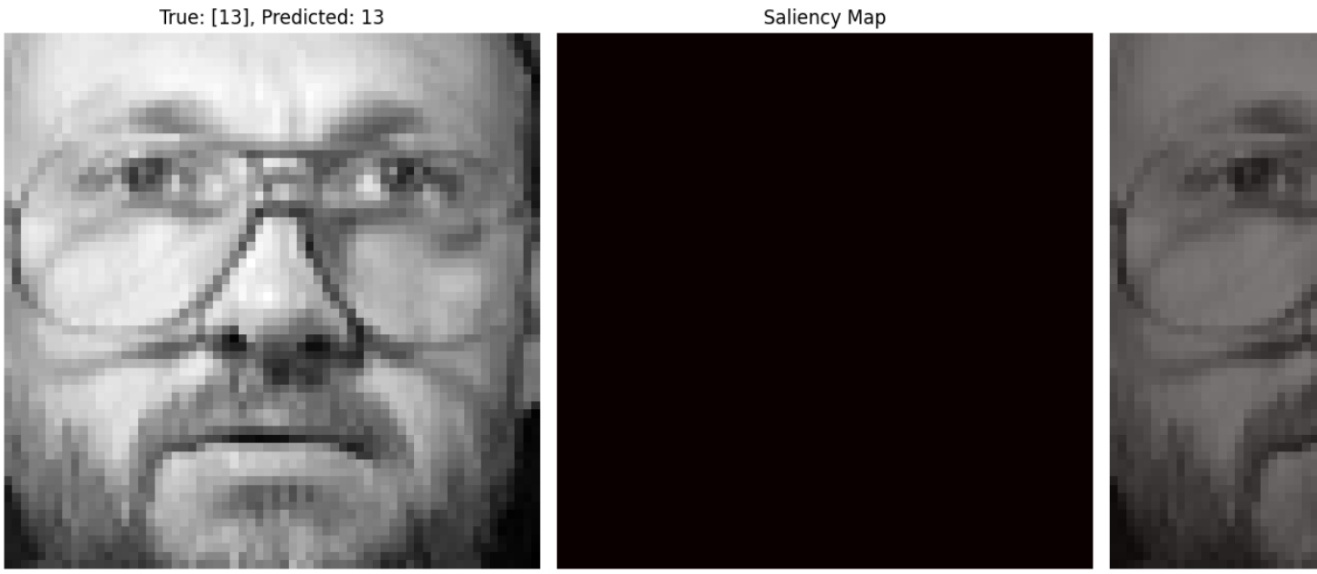


Figure 9: Saliency map for the predictions of glassbox models

of the image that most influenced the model's decision, highlighting important features and providing an intuitive understanding of the model's behavior.

8. Emerging Trends and techniques Discussion:

In our pursuit of improving model performance, we turned to the emerging technique of transfer learning. Initially, the custom model for our Siamese neural network did not yield satisfactory results. To address this, we incorporated VGG19 as the backbone for the Siamese network, leveraging its pre-trained ImageNet weights.
We began by converting our grayscale images to RGB format to match the input requirements of VGG19. Using the converted images, we created pairs and corresponding labels, which were then split into training and testing sets. We loaded VGG19 without the top layer and froze its pre-trained layers to maintain the valuable feature representations learned from the extensive ImageNet dataset.
To tailor the model for our specific task, we added a custom top layer. This included flattening the output of VGG19 and adding a dense layer with 256 neurons and ReLU activation. By combining the robustness of VGG19's pre-trained features with our custom top layer, we aimed to significantly enhance the model's performance on our dataset.
By using this method, the performance improved significantly, showcasing the effectiveness of transfer learning. The use of transfer learning in this context exemplifies a powerful emerging trend in machine learning, where leveraging pre-trained models can lead to substantial improvements in performance, especially when dealing with limited data. This approach allowed us to overcome the limitations of our initial custom model and achieve better results.

9.Ethical Considerations

Regardless of the underlying model architecture, be it Siamese networks or K-Nearest Neighbors (KNN), These systems have the potential to impact individuals' rights to privacy, fairness, and autonomy. One ethical concern revolves around privacy, as face ver-

ification systems inherently involve the processing and storage of individuals' biometric data, which raises concerns about unauthorized access, misuse, and potential breaches. Additionally, there is a risk of bias and discrimination, particularly in KNN models where the choice of distance metric and the representation of training data can inadvertently reinforce existing societal biases. Moreover, the deployment of face verification systems in various contexts, such as law enforcement or employment screening, raises questions about fairness and potential discrimination based on facial characteristics. Therefore, it is imperative for developers and stakeholders to prioritize transparency, accountability, and fairness throughout the design, development, and deployment of face verification systems, ensuring that they uphold ethical standards and respect individuals' rights and dignity.

10.Legal and Regulatory Compliance

In the domain of face verification and authentication systems, legal and regulatory compliance is crucial to ensure the protection of individuals' rights, privacy, and security. Several laws and regulations govern the use of biometric data, including facial recognition technology, to mitigate potential risks and ensure responsible deployment.

One significant legal framework is the General Data Protection Regulation (GDPR) in the European Union (EU), which establishes strict requirements for the processing of personal data, including biometric information. Under the GDPR, facial recognition systems must adhere to principles of lawfulness, fairness, and transparency. Individuals must provide explicit consent for the collection and processing of their biometric data, and organizations must implement robust security measures to safeguard this sensitive information.

In addition to the GDPR, various countries and regions have enacted specific laws and regulations governing the use of facial recognition technology. For example, the California Consumer Privacy Act (CCPA) and the Biometric Information Privacy Act (BIPA) in the United States impose requirements for the collection, storage, and use of biometric data, including facial images. These laws often mandate informed consent, data protection measures, and provisions for individuals to access and control their biometric information.

Furthermore, regulatory bodies such as the U.S. Federal Trade Commission (FTC) and the Information Commissioner's Office (ICO) in the UK actively monitor and enforce compliance with data protection regulations. They have the authority to investigate incidents of data breaches, unauthorized data sharing, or misuse of biometric information and impose significant fines and penalties on non-compliant organizations.

Moreover, emerging legislation and guidelines, such as the proposed EU Artificial Intelligence Act and the OECD Principles on Artificial Intelligence, aim to address the ethical and societal implications of facial recognition technology. These frameworks emphasize principles of accountability, transparency, and fairness in the development and deployment of AI systems, including those used for face verification and authentication.

In summary, legal and regulatory compliance is paramount in the field of face verification and authentication systems to ensure the ethical and responsible use of biometric data. Organizations must stay abreast of evolving laws and regulations, implement robust data protection measures, and prioritize the rights and privacy of individuals in their deployment of facial recognition technology.

11.Summary and Conclusion

In conclusion, the comparison between the Blackbox (Siamese network) and Whitebox (KNN) models for facial recognition using the Olivetti dataset yielded valuable insights into their respective strengths and limitations. While the Siamese network focused on learning similarities between data points, KNN measured similarity between feature vectors. The study revealed that the Siamese network with a VGG19 backbone achieved superior accuracy of around 95 percentage showcasing the advantages of leveraging pretrained networks for facial recognition tasks. XAI visualization with Grad CAM further facilitated the interpretation of the Siamese network's predictions, enhancing transparency and interpretability. However, it was noted that the Siamese neural network required extensive fine-tuning and layer configurations for optimal performance, particularly when using custom models. Despite this, its potential to capture complex datasets makes it a promising option for facial recognition tasks. On the other hand, KNN offered simplicity and ease of implementation with satisfactory accuracy of 91.25 percentage. In summary, the study emphasized the importance of selecting the appropriate model based on task requirements and dataset characteristics, acknowledging the trade-offs between complexity, accuracy, and implementation efforts.

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